

# FALCON: LandslideGuard AI

## An Easy-to-Understand Guide to How the Landslide Detection & Early Warning System Works

**Target Audience:** Project Stakeholders, Disaster Management Teams & Evaluators | **Team:** Syntax Terrors

### 1. The Big Picture: What Problem Does FALCON Solve?

In hilly and mountainous areas, heavy rain, steep slopes, and unstable soil can cause sudden, devastating **landslides**. Landslides destroy roads, bury homes, block vital transportation routes, and claim human lives with little to no advance warning.

Traditionally, people monitor slopes by sending human inspectors to climb dangerous hills or by relying on simple rain gauges. These traditional methods are either too slow, too dangerous, or fail to catch the warning signs before the mountain gives way.

**FALCON LandslideGuard AI is an automated 24/7 intelligent guardian.** It acts like an expert team of scientists, drone pilots, and sensors working together in real time. It constantly looks at hillsides from the sky, checks satellite images from space, measures ground moisture and vibrations, and forecasts upcoming rainstorms to give communities hours of early warning before a disaster happens.

### 2. How Does FALCON Work? (The 4 Eyes of the System)

FALCON does not rely on just one source of information. Instead, it combines four different layers of data to be completely sure of what is happening on the ground:

| The 4 Pillars                          | How It Collects Data                                                                              | What It Spots (In Simple Terms)                                                                                                              |
|----------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Drones (The Eyes in the Sky)        | Flying drones with high-definition cameras capture live aerial photos of the hillsides.           | Artificial Intelligence looks at every pixel of the video to spot newly cracked earth, falling rocks, bare soil patches, and sliding ground. |
| 2. Satellites (The View from Space)    | Earth-observation satellites take multi-color (multispectral & infrared) scans of entire valleys. | Identifies hidden moisture across square kilometers, tree cover loss, and large-scale shifts in mountain rock that human eyes cannot see.    |
| 3. Ground Sensors (Nerves in the Soil) | Small electronic sensor boxes (ESP32 microcontrollers) installed directly in the hillside soil.   | Measures underground water pressure, soil wetness, temperature, air quality, and tiny millimeter-scale ground vibrations.                    |
| 4. Live Weather (The Future Outlook)   | Connected to live meteorological radar and forecast services (WeatherAPI).                        | Forecasts rainstorms 6 hours and 24 hours in advance to know when soil will become too heavy and water-logged to stay attached.              |

### 3. The 'Brain' of FALCON: The 4-Tier Smart Risk Engine

Instead of just averaging numbers together, FALCON uses a 4-tier decision process based on actual mountain physics to ensure total accuracy without false alarms:

- **Tier 1 — The Slope Check (Physics First):** Water on flat ground causes puddles, not landslides! If the land is flat (less than 15 degrees steep), the system automatically prevents false alarms. A steep mountain slope is required before danger levels rise.
- **Tier 2 — The AI Machine Learning Calculator:** An advanced mathematical model (XGBoost) studies the exact mix of soil wetness, underground pressure, slope angle, and rain to calculate an initial hazard score from 0 to 100.
- **Tier 3 — The Emergency Safety Override (The Panic Button):** If ground movement sensors detect that the soil has physically shifted by **8 millimeters or more**, or if underground water pressure reaches dangerous levels, FALCON immediately overrides all calculations and triggers a **CRITICAL EMERGENCY** alarm.
- **Tier 4 — Visual Double-Checking (Preventing Fake Alarms):** If a drone sees what looks like a landslide, the system checks whether the same slip is visible across multiple video frames. A momentary shadow or a passing car won't cause an alarm, but a confirmed visual slip instantly boosts the danger rating.

## 4. The Traffic Light Danger Scale (What the Scores Mean)

FALCON converts complex sensor math into a clear, intuitive 4-color traffic light system so emergency operators can act immediately:

| Score    | Risk Level | Visual Color | What It Means in Real Life                                             | Recommended Action                                                                |
|----------|------------|--------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 0 – 24   | SAFE       | GREEN        | Stable hillside; normal moisture and no ground movement.               | Routine automated monitoring; no action needed.                                   |
| 25 – 49  | MODERATE   | YELLOW       | Rain is falling; ground moisture and water pressure are increasing.    | Send advisory alerts; increase drone patrol frequency.                            |
| 50 – 74  | HIGH       | ORANGE       | Soil is heavily saturated; minor earth shifts detected on steep slope. | Alert emergency response teams; prepare road barricades and warnings.             |
| 75 – 100 | CRITICAL   | RED          | Imminent slope collapse! Active ground displacement or massive scarp.  | <b>IMMEDIATE EVACUATION!</b> Sound public sirens, close roads, dispatch rescuers. |

## 5. What Does the User See? (The Command Dashboard)

FALCON comes with a modern, easy-to-use web dashboard designed for disaster management control rooms, field workers, and government authorities:

- **Live Drone Video Monitor:** Displays the real-time camera feed from flying drones. If a landslide crack appears, the AI automatically draws a bright orange highlight box around the hazard area with a percentage breakdown.
- **Interactive Geographic Map:** Shows an OpenStreetMap view of the region, pinpointing exact GPS coordinates of the drone, nearby towns, mountain ridges, and active alert spots.
- **'What-If' Simulation Laboratory:** Emergency planners can drag virtual sliders for rainfall, soil wetness, ground movement, and slope angle to simulate worst-case monsoon scenarios and test town safety plans.
- **Incident Snapshot Library:** Automatically saves photo evidence, sensor records, and AI confidence reports into a searchable history log for disaster auditing and post-event analysis.

## 6. Key Benefits & Real-World Impact

- **Saves Lives & Reduces Casualties:** Gives authorities hours of advance warning to evacuate vulnerable hillside communities before a disaster occurs.
- **Prevents False Alarms:** By cross-referencing drone photos with underground sensors and slope physics, FALCON avoids false panic.
- **Operates 24/7 in Real Time:** Ground sensors and weather monitors work continuously, updating danger assessments every two seconds.
- **Cost-Effective & Scalable:** Uses lightweight, affordable IoT microcontrollers and standard drones, making it deployable across entire mountain ranges.